

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

Computer Aided Design (ME751)

INDEX

Sr. No.	Date	Title	No. of Pages	Marks	Date of Assessment	Sign of Faculty
1		Introduction to CAD and Graphics Hardware				
2		Study and Application of Computer Graphics in CAD				
3		Algorithms Used for Generating 2D Output Primitives				
4		Geometric Transformations				
5		Design of Machine Elements				
6		Geometric Modeling (Part Modeling)				
7		Geometric Modeling (Assembly Modeling)				
8		Mathematical Elements of Curves				
9		Reverse Engineering				
10		Case-Study on Applications of CAD				

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

Computer Aided Design (ME751)

Date:

EXPERIMENT NO. 1
INTRODUCTION TO CAD AND GRAPHICS HARDWARE

AIM:

Introduction to CAD and Graphics Hardware

OBJECTIVES:

1. Introduction to CAD and its role in the industry.
2. To know the requirement of software & graphics hardware in CAD system.

THEORY:

Computer Aided Design can be defined as the application of a computer system to a solution of design problems. It involves any type of design activity, which makes use of the computer to develop, analyze, or modify an engineering design. In modern CAD systems ICG is used. The interactive computer graphics denotes a user-oriented system in which the computer is employed to create, transform and display data in the form of pictures or symbols.

The main reasons for implementing CAD system are:

- Increase in the productivity of the designer.
- Improvement in the quality of design.
- Improves communication
- Increase in efficiency of design.
- Strengthen the manufacturing database.

QUESTIONS:

1. List out the hardware and software you will go for establishing a CAD center giving full latest technical specification for a CAD work station.
2. Using a simplified description of process of design explain where computational aids might be assistance to the designer.
3. Distinguish between Passive & Interactive Graphics.
4. Describe the capabilities of a typical 3D – CAD software.

REFERENCES:

1. CAD/CAM Theory and Practice by Ibrahim Zeid
2. Computer Graphics and Design by P Radhakrishan and C P Kothandramen

Marks obtained:	Signature of faculty:	Date:
------------------------	------------------------------	--------------

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

Computer Aided Design (ME751)

Date:

EXPERIMENT NO. 2
STUDY AND APPLICATION OF COMPUTER GRAPHICS IN
CAD

AIM:

Study of computer graphics in CAD.

OBJECTIVES:

1. Review of graphics functions in C language.
2. Generation of 2D entities using C language.

THEORY:

Computer graphics can be defined as creation, storage and manipulation of pictures and drawings by means of a computer. It enhances the power of communication between the computer and its users. In a graphic mode, it is possible to display graphical figures and text. The graphic mode uses the pixel, which depends on resolution of monitor. The graphic functions are available in C language in a file 'graphics.h' which must be included in the program.

The graphics screen can be initiated by specifying the graphics driver and mode or by automatic detection. Automatic detection is the best way of initiating graphics screen and can be achieved by the following command:

graphdriver = DETECT
initgraph (&*graphdriver*, &*graphmode*, " ");

where *graphdriver* and *graphmode* are integer data types. The graphics screen is closed by the command *closegraph* ();

When we switch over to graphics mode, cursor disappears and a coordinate system is established whereby the top left corner of the screen is treated as origin (0, 0). C++ is a powerful language for graphics and there are several routines available to create graphic image.

QUESTIONS:

1. Enlist & explain various commands/functions of graphics mode operation on your system.
2. Explain ready-made function available for drawing a point, line, arc, circle & ellipse, for setting different line styles & to write text on your system.
3. Write a computer program to draw a rectangle of sides 300 and 200 units and four circles within it, each circle touching two adjacent sides of rectangle. Assume suitable diameter of the circles.

REFERENCES:

1. Computer Aided Design by Dr. S S Khandare
2. Computer Graphics and Design by P Radhakrishnan and C P Kothandramen

Marks obtained:	Signature of faculty:	Date:
------------------------	------------------------------	--------------

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

Computer Aided Design (ME751)

Date:

EXPERIMENT NO. 3
ALGORITHMS USED FOR GENERATING
2D OUTPUT PRIMITIVES

AIM:

To study routines for generating 2D output primitives.

OBJECTIVE:

Study of algorithms for displaying line and circle using C graphics/MATLAB.

THEORY:

Routines/algorithms for generating output primitives provide the basic tools for constructing pictures. Each output primitive is specified with input coordinate data and other data about the way that object is to be displayed. Points and straight-line segments are the simplest geometric components of pictures. We will study device level algorithms for displaying 2D output primitives, with particular emphasis on scan conversion methods for raster graphics system.

Line drawing is accomplished by calculating intermediate positions along the line path between two specified end point positions. An output device is then directed to fill in these positions between the end points. The process of turning on the pixels for a line segment is called 'vector generation'.

Circle is a frequently use component in pictures and graphs, a procedure for generating either full circle or circular arcs is included in most graphics packages.

QUESTIONS:

1. Draw a flow chart & prepare a computer program for scan conversion of a line using DDA algorithm.
2. Write a program to generate line with endpoints (5, 5) & (12, 15) using Bresenham's algorithm.
3. Write an algorithm to draw circle using Bresenham's method, also sketch the pixels for representing a circle with center at origin and radius = 10 units.
4. Draw a flow chart & prepare a computer program to scan convert an ellipse. Use Bresenham's method of scan conversion.

REFERENCES:

1. Computer Graphics by Donald Hearn and M Pauline Baker
2. Computer Graphics and Design by P Radhakrishnan and C P Kothandramen
3. Computer Aided Design by Dr. S S Khandare

Marks obtained:	Signature of faculty:	Date:
------------------------	------------------------------	--------------

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

Computer Aided Design (ME751)

Date:

EXPERIMENT NO. 4
GEOMETRIC TRANSFORMATIONS

AIM:

To study transformation of geometric models.

OBJECTIVE:

To simulate geometric transformation of 2D entities using programming language.

THEORY:

Computer graphics image are generated from a series of line segments that are represented by the coordinates of their end points. In many applications there is need for altering or manipulating displays. Certain changes in an image can be easily made by performing mathematical operation on these coordinates, as image has been coded as numbers and stored in computer.

Geometric transformation provides a means by which an image can be enlarged in size, or reduced, rotated or moved. These changes are brought about by changing the coordinates of the image to a new set of values depending upon the requirement. Basic geometric transformations are Translation, Rotation & Scaling and other transformations include Reflection and Shearing.

QUESTIONS:

1. Write program to scale, translate, rotate & mirror the given line about origin.
2. Write program to translate and rotate triangle about origin.
3. Write a program to shear a rectangle in x & y directions.

REFERENCES:

1. Computer Aided Design by Dr. S S Khandare
2. Computer Graphics and Design by P Radhakrishan and C P Kothandramen
3. Computer Graphics by Donald Hearn and M Pauline Baker

Marks obtained:	Signature of faculty:	Date:
------------------------	------------------------------	--------------

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

Computer Aided Design (ME751)

Date:

EXPERIMENT NO. 5
DESIGN OF MACHINE ELEMENTS

AIM:

Computer aided design of machine elements.

OBJECTIVE:

To design machine elements using any programming language.

THEORY:

CAD is useful for design of structural components, machine parts of complex shapes and nature. This allows preparation of simple program to calculate design dimensions and provides a drawing of the design using computer graphics.

The design of machine components generally involves the calculations of stress, strain or dimensions of component. It is necessary in analysis or design program to know the type of input and output variables.

QUESTIONS:

1. Write a computer program for four bar chain mechanism PQRS for determination of angular displacements and angular velocities of links QR & RS at an interval of θ of the input link PQ. PS is fixed link.
2. For a connecting rod of an IC engine write an interactive program.
3. Write an interactive program for the design of spur gear.
4. Prepare an exhaustive algorithm for:
 - a) Design of shaft subjected to no. of point loads in one plane and transmitting power from any one given point to some other point on the shaft.
 - b) Design of helical compression springs
 - c) Design of spindle of a lathe

REFERENCE:

- Design of Machine Elements by Sharma Purohit
- Computer Aided Design by Dr. S S Khandare
- PSG Design Data Book

Marks obtained:	Signature of faculty:	Date:
------------------------	------------------------------	--------------

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

Computer Aided Design (ME751)

Date:

EXPERIMENT NO. 6
GEOMETRIC MODELING (PART MODELING)

AIM:

To study different geometric modeling techniques.

OBJECTIVES:

1. Study of various representation techniques for solid & surface modeling.
2. To create geometric models by using drafting & modeling software.

THEORY:

Computer representation of the geometry of component using software is called a geometric model. Geometric modeling is a process of generating 3D objects for analysis, design, drafting and manufacturing. It creates database, which is used to display the object, to prepare data for design and analysis, also to manufacture the objects using CNC machines.

Geometric modeling is done in three principal ways. They are:

- a) Wire frame modeling
- b) Surface modeling
- c) Solid modeling

QUESTIONS:

1. Explain various representation schemes for solid and surface modeling.
2. Explain feature based models stating advantages and limitations. How are those constructed?
3. What is parametric modeling?
4. List AutoCAD commands for following:
 - a) Wire frame entities
 - b) Surface entities
 - c) Solid primitives
 - d) Boolean operations used for solid modeling
5. Prepare solid model of a given object with components and complete assembly and the drawing view using Pro/E Wildfire 5.0.

REFERENCES:

1. Computer Aided Design by Dr. S S Khandare
2. CAD/CAM/CIM by Radhakrishnan, Subramanyan & Raju
3. Pro/E Tutorial

Marks obtained:	Signature of faculty:	Date:
------------------------	------------------------------	--------------

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

Computer Aided Design (ME751)

Date:

EXPERIMENT NO. 8
MATHEMATICAL ELEMENTS OF CURVES

AIM:

To study mathematical elements of Curves for geometric modeling.

OBJECTIVE:

1. To study differential geometry of curves.
2. To study various representation and design of curves.

THEORY:

The ultimate goal of geometric modeling is to create solid models which are constructed by points, lines and curves. We extend curves to create surfaces and then use surfaces to create solid. Thus, CAD designers must master curves concepts before they can create complex solid models.

There are two categories of curves that can be represented parametrically: analytic and synthetic. Analytic curves are defined as those that can describe by analytic equations such as lines, circles and conics. Synthetic curves are ones that are described by sets of data points (control points) such as splines and Bezier curves. Thus a designer should know the mathematical properties of each entity and how it is used in geometric modeling.

Splines are used in graphics applications to design curve and surface shapes, to digitize drawings for computer storage, and to specify animation paths for the objects or the camera in a scene. Typical CAD applications for splines include the design of automobile bodies, aircraft and spacecraft surfaces, and ship hulls.

QUESTIONS:

1. List various properties of Bezier Curves.
2. Write short notes on B-spline curve & NURBS.
3. The coordinates of four control points relative to a current WCS are given by $P_0 = [2 \ 2 \ 0]^T$, $P_1 = [2 \ 3 \ 0]^T$, $P_2 = [3 \ 3 \ 0]^T$ & $P_3 = [3 \ 2 \ 0]^T$. Find the equation of the resulting Bezier curve. Also find points on curve for $u = 0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, \text{ \& } 1$.

REFERENCES:

1. Mathematical Elements for Computer Graphics by Rogers and Adams
2. Mastering CAD/CAM by Ibrahim Zeid

Marks obtained:	Signature of faculty:	Date:
------------------------	------------------------------	--------------

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

Computer Aided Design (ME751)

Date:

EXPERIMENT NO. 7
GEOMETRIC MODELING (ASSEMBLY MODELING)

AIM:

To create assembly and drawing using 3D geometric modeling software

OBJECTIVES:

1. Create and modify Assembly and details.
2. Create drawings that display views of multiple models.

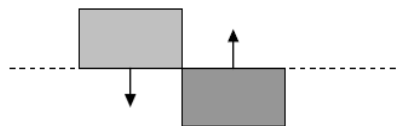
THEORY:

An assembly is a collection of components (parts and sub-assemblies) arranged in a specific way. Assembly modeling is technology and methods used by computer-aided design and product visualization computer software systems to handle multiple files that represent components within a product. The components within an assembly are represented as solid or surface models. The individual data files describing the 3D geometry of individual components are assembled together through a number of sub-assembly levels to create an assembly describing the whole product.

To assembly two components is equivalent to constrain all 6 DOF's between the two. There are six types of common constraints those are described below.

MATE (or MATE COINCIDENT)

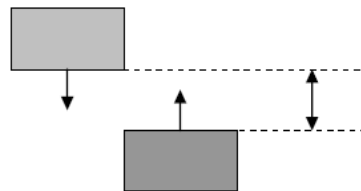
Two planar surfaces or datums become coplanar and face in opposite directions.



MATE constraint.

MATE OFFSET

Two planar surfaces or datums are made parallel, with a specified offset distance, and face in opposite directions. The offset dimension can be negative.

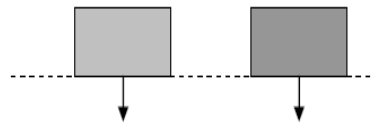


MATE OFFSET constraint.

ALIGN (or ALIGN COINCIDENT)

This can be applied to planar surfaces, datums, revolved surfaces and axes. Planar surfaces become coplanar and face in the same direction.

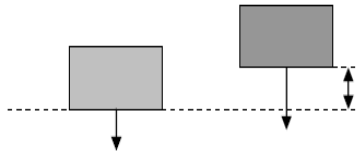
CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering



ALIGN constraint with planar surfaces

ALIGN OFFSET

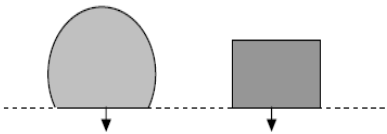
This can be done only with planar surfaces: they become parallel with a specified offset and face the same direction.



ALIGN OFFSET constraint.

ALIGN ORIENT

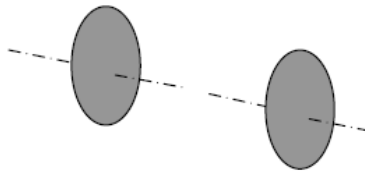
Two planar surfaces or datums are made parallel and face the same direction (similar to Align Offset except without the specified offset distance).



ALIGN ORIENT constraint.

INSERT

This constraint can only be used with two surfaces of revolution in order to make them coaxial.



INSERT constraint.

QUESTIONS:

1. Prepare Assembly model using minimum 5-6 solid model components and the relevant drawing views using modeling software.

REFERENCES:

1. Computer Aided Design by Dr. S S Khandare
2. CAD/CAM/CIM by Radhakrishnan, Subramanyan & Raju
3. Pro/E Tutorial

Marks obtained:

Signature of faculty:

Date:

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

Computer Aided Design (ME751)

Date:

EXPERIMENT NO. 09
REVERSE ENGINEERING

AIM:

Introduction to reverse engineering and its scope in CAD

OBJECTIVES:

1. To study about different types of 3-D scanners for reverse engineering.
2. To study different types of digitizing (scanning) techniques for reverse engineering.
3. To learnt how to generate a 3-D CAD model by using XOR/Redesign software.

THEORY:

Engineering is the profession of designing, manufacturing, constructing, and maintaining. In today's world of competition, we always look for the best, the most efficient, and the most durable to meet the current needs. These requirements led to the development of reverse engineering phenomenon.

Reverse Engineering, is a process that is used to create 3D CAD models directly from physical parts with little or no additional design documentation. By scanning (digitizing) items that have insufficient design documentation or require CAD data for modern manufacturing methods, a component can be quickly and accurately reverse engineered.

QUESTIONS:

1. What is reverse engineering? Justify its use.
2. Explain in brief different type of scanners? What are active scanner and passive scanner?
3. Write down specification of 3D scanner (LPX-60 model).
4. List out different reverse engineering software.
5. What are Key Benefits of Rapidform XOR/Redesign?
6. Practice a simple 3-D CAD model by using RE setup facility in the laboratory.

REFERENCES:

1. Reverse Engg. Set-up (MODROB-AICTE)
2. <http://www.rolanddga.com/asd/products/scanners/LPX60/>
3. <http://www.npd-solutions.com/reoverview.html>
4. Reversing: Secrets of Reverse Engineering by Eldad Eilamv. Publisher: Wiley
5. Rapidform XOR/Redesign tutorials.
6. <http://nptel.iitm.ac.in/courses/Webcourse-contents/IIT-Delhi/ComputerAidedDesignManufacturingII>

Marks obtained:	Signature of faculty:	Date:
------------------------	------------------------------	--------------

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
CHAMOS Matrusanstha Department of Mechanical Engineering

Computer Aided Design (ME751)

Date:

EXPERIMENT NO. 10
CASE-STUDY ON APPLICATIONS OF CAD

AIM:

To explore the applications of CAD in different areas of engineering.

OBJECTIVE:

To understand the scope and utility of different CAD based software's in engineering field applications.

THEORY:

CAD/CAM has been utilized in engineering practice in many ways including drafting, design, simulation, analysis, and manufacturing. There are considerable numbers of applications for the various existing CAD/CAM systems. Each application has its own strengths and is usually targeted towards a specific market and groups of users. For example, there are mechanical, electrical, and architectural CAD and CAM products.

An inspection of these various systems reveals that they have generic structure and common modules. Awareness of such structures and modules enables users to better understand system functions for both evolution and training purposes.

QUESTIONS:

1. Search on the internet or develop on your own a CAD based application in engineering field and discuss the scope of it as a part of your presentation.
 - a. The topic should be relevant and feasible as a project to be undertaken by the student subsequently.
 - b. It can be in the form of a white paper, journal paper, idea or concept put in a presentable form.
 - c. The student should present it in the form of a report discussing the objective, scope and procedure to do the case-study and conclude the findings of the same.

REFERENCES:

1. Research papers
2. Magazines & Periodicals

Marks obtained:	Signature of faculty:	Date:
------------------------	------------------------------	--------------